

Notes: Conditionals

CS 101 Algorithmic Problem Solving

Fall 2024

Need for Conditionals

Conditional statements are features of programming languages that tell the computer to execute certain actions, provided that certain conditions are met. The decisions made are if and only if the pre-stated conditions are either true or false, depending on what the programmer has decided.

if statements

The **if** statement is used to select actions to perform. The **if** statement may contain other statements, including other **ifs**. Those are known as nested **ifs** and would be seen later. The **if** statement is used like a test, it can be followed by one or more optional **elif** (else if) tests and a final optional **else** block.

General format

followed by text represents a comment.

```
if test1:           #if test
    statement1      #Associated block
elif test2:         #optional elif
    statement2
else:               #optional else
    statement3
```

Examples

example 1

```
a = 3
if a == 3:
    print(a)
else:
    print("not three")
```

example 2

```
name = "ali"
if name == "ali":
    print("Hello ali")
elif name == "raahim":
    print("Hello raahim")
elif name == "":
    print("No name")
else:
    print("Why are we doing this?")
```

example 3

```
var = 100

# We have an initial if where the value of var is checked,
# if it is less than 200 we proceed inside code block

if var < 200:
    print "Expression value is less than 200"

    # Here we have another if condition, notice that this if condition
    # is inside the first one making it a nested if.

    if var == 150:
        print("Which is 150")
    elif var == 100:
        print("Which is 100")
    elif var == 50:
        print "Which is 50"
    elif var < 50:
        print "Expression value is less than 50"

# This else corresponds to the first if condition and is executed in the scenario
# where the value is greater than 200.

else:
    print "Could not find true expression"
```

Boolean Arithmetic

The **if** statement takes action on *truth values* (**True** or **False**) that are usually produced through some tests. *Boolean operators* are used to combine the results of multiple tests to produce new truth values.

The three Boolean expression operators we will look at are: **and**, **or**, and **not**.

Pseudocode

```
# assuming X and Y are Boolean variables, i.e. the value of each is True or False
X and Y # is True if both X and Y are True, otherwise it is False.
X or Y # is True if any of X or Y is True, otherwise it is False.
not X # is True if X is False, otherwise it is False.
```

Example

```
A = True
B = False

A or B # evaluates to True as one of the operands is True
A and B # evaluates to False as not both operands are True
not B # evaluates to True as the operand is False
not A # evaluates to False as the operand is True
A == B # evaluates to False as the operands have different values
A != B # evaluates to True as the operands have different values
```